Amrit Gopinath

B.E. Computer Science & Engineering · AI/ML Research

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Education

Sri Sivasubramaniya Nadar College of Engineering, Chennai 2024–2028 (expected)
B.E. Computer Science and Engineering; Honors track in AI/ML.
CGPA: 9.383/10.0 | **Department rank: 1st** (through Semester 4).

Research Interests

Multilingual language models, expert routing, and representation analysis; causal interpretability and model introspection, with interests in vision–language models and AI safety.

Selected Research

A Declarative–Procedural Perspective on Expert Routing in Bilingual Mixture-of-Experts Language Models 2026
First author | ARR 2026 submission | arXiv:2608.15102 | Code

- Investigated whether bilingual MoE routing captures lexical, grammatical, and syntactic structure, or mainly reflects language identity and frequency.
- Led the design of controlled English–German corpora, curriculum and mixed-language training, and held-out linguistic probes; analyzed routing with mutual information, entropy, Jensen–Shannon distance, frequency controls, and routing ablations.
- Peak routing-category MI was **0.2599 without a curriculum** versus **0.1148 with a curriculum**. Replication across **2 seeds** exposed seed-dependent language dominance without a curriculum; curriculum routing was more language-balanced.

Having a State Is Not Knowing It 2026
Co-author | Apart Research Digital Minds Research Sprint | Report | Project

- Examined whether language models can report internal decision states and predict how causal interventions change their behavior.
- Replicated Llama concept injection and built a controlled Qwen benchmark with measured logit-margin ground truth, matched external probes, separate counterfactual prediction/scoring passes, and shortcut controls.
- Self-report improved from **0.243 to 0.522 macro-F1**; a matched external MLP reached **1.000 F1** on the controlled benchmark. Intervention-direction F1 was **0.695 on 96 held-out trials**, falling to **0.564** on unseen mixtures; quantitative response prediction had negative R^2.

Research Experience

NLP Research Intern, National Institute of Technology Tiruchirappalli 2026
Faculty mentorship | First-author manuscript submitted to ICON 2026 | Paper | Code

- Studied what neural models can recover from scarce Classical Tamil verse–commentary data beyond lexical overlap and source-specific patterns.
- Used **1,262 pairs** to compare recurrent and Transformer encoders, Siamese matching, and generation models against lexical, random-data, and constant-output controls, with held-out and out-of-domain evaluation.
- The decoder preferred authentic word order in **107/112 comparisons (95.5%)**. On **30 held-out rows**, generation scored **0.060 token-F1** versus **0.108** for a constant-word baseline, exposing limited verse-specific generation.

Publications and Benchmark Research

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection 2026

MediaEval 2026 Working Notes | First author | Paper | Code

- Investigated synthetic-image detection across diverse generators and the effects of training-data diversity, input representation, and model architecture.
- Compared Swin, global and MIL-style aggregation, luminance-only inputs, and CNN/ResNet–GRU–ViT hybrids under constrained and open training settings.
- Swin Base achieved **0.6997 synthetic-class F1** and **0.7471 AUC** on **open validation**, using SID_Set as external training data.

FinMMEval 2026: TextSentinels 2026

CLEF 2026 Working Notes (CEUR-WS, Vol. 4283) | 3 published task papers | Co-author

- Studied multilingual financial question answering from heterogeneous evidence and the use of financial text for trading decisions across multiple assets.
- Contributed to FAISS retrieval with question-type model routing and option scoring, BM25 retrieval preserving SEC tables and multilingual evidence, and FinBERT/XGBoost trading with recursive sentiment memory.
- Final MCQA accuracy was **81.5% / 76.5% / 86.0%** for English/Chinese/Arabic (**200 questions per language**). Financial-QA ROUGE-1 F1 improved from approximately **0.04 to 0.28 locally** (approximately **0.18** on the final benchmark); historical backtests across **6 assets** yielded a **mean per-asset annualized Sharpe of 2.03**.

Published task papers (titles below share the prefix "TextSentinels @ FinMMEval 2026"):

Task 1: A Multilingual Routed Retrieval-Augmented System for Financial Multiple Choice Questions. pp. 2635–2643.

Task 2: Chronological Stabilization of a BM25-Grounded Multilingual Financial Question Answering Pipeline. pp. 2644–2651.

Task 3: Multi-Asset Financial Decision Making Using Machine Learning and Financial Transformers. pp. 2624–2634.

Technical Skills

ML / NLP	PyTorch, Hugging Face Transformers, scikit-learn, FAISS, Stanza.
Research methods	Mutual and conditional mutual information, routing entropy, Jensen–Shannon distance, permutation tests, concept injection, external probes, counterfactual interventions; CCA with matched null controls.
Programming / tools	Python, C, Java, TypeScript, Git, Linux, LaTeX.

Academic Service

- **Reviewing assistance, CLEF 2026 Working Notes:** assessed methodological clarity, experimental validity, presentation, and reproducibility.